

CAP182 SS2

Part A

Title	Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns
Harvard Reference	Mishra, A. and Dutta, P. (2024) 'Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns', <i>Journal of Cleaner Production</i> , 477, 143802. doi: 10.1016/j.jclepro.2024.143802.
Link	https://www.sciencedirect.com/science/article/abs/pii/S0959652624032517?
Modelling	Logistic Regression, Random Forest and Gradient Boosting. The models were evaluated using ROC curves and AUC. Gradient Boosting produced the strongest predictive performance in the study.
Preprocessing Techniques	The study prepared operational e-commerce data containing order, product and transaction characteristics for predictive modelling. Relevant variables included order quantity, discount, total price, product category, payment mode and shipping-related variables.
Datasets Used	Data from an Indian e-commerce company selling industrial equipment. The observations covered approximately three months, from July 2017 to October 2017, with 38 product categories.
Key Takeaways	1. Machine learning can be used to predict e-commerce product returns. 2. Order and product characteristics can provide useful predictors of return behaviour. 3. Model evaluation should go beyond simple accuracy and consider measures such as ROC-AUC.

Title	A Local Algorithm for Product Return Prediction in E-Commerce
Harvard Reference	Zhu, Y., Li, J., He, J., Quanz, B.L. and Deshpande, A.A. (2018) 'A Local Algorithm for Product Return Prediction in E-Commerce', <i>Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence</i> , pp. 3718–3724. doi: 10.24963/ijcai.2018/517.
Link	https://www.ijcai.org/proceedings/2018/517
Modelling	A local algorithm designed specifically for large-scale product-return prediction. The approach incorporates customer behaviour, product information and purchase/return history.
Preprocessing Techniques	The approach makes use of customer purchase behaviour, previous returns, product purchase history and customer/product similarity. The paper focuses particularly on incorporating historical behavioural information into return prediction.
Datasets Used	Large-scale e-commerce data involving customers, products, purchases and returns.
Key Takeaways	1. Historical customer return behaviour can be valuable when predicting future returns. 2. Product and customer information can be combined rather than treating each transaction independently. 3. Large e-commerce datasets require modelling approaches that can handle substantial numbers of customers and products.

Title	Predicting product return volume using machine learning methods
Harvard Reference	Cui, H., Rajagopalan, S. and Ward, A.R. (2020) 'Predicting product return volume using machine learning methods', <i>European Journal of Operational Research</i> , 281(3), pp. 612–627. doi: 10.1016/j.ejor.2019.05.046.
Link	https://www.sciencedirect.com/science/article/abs/pii/S037722171930517X?
Modelling	Multiple machine-learning approaches were investigated, including main-effects and interaction-effects models. LASSO was particularly useful because it performed variable selection while modelling many possible interactions.
Preprocessing Techniques	The researchers constructed predictive variables from sales, time, retailer, product type and detailed operational information. Interaction effects were also considered.
Datasets Used	A detailed operational dataset from a US manufacturer covering products sold and/or returned over 39 months.
Key Takeaways	1. Historical sales and return information can help forecast future returns. 2. Feature selection can be important when there are many possible predictors and interactions. 3. Return prediction can support operational decisions such as inventory and returns management.

Name	E-Commerce Returns: Who Will Send It Back?
Harvard Reference	Kaggle (2026) <i>E-Commerce Returns: Who Will Send It Back?</i> Kaggle.
Link	https://www.kaggle.com/competitions/retail-return-risk-modeling/data?
No. of Instances	There are a total of 250000 records in the dataset, with 200000 present in the training data and 50000 in a separate test dataset (unusable for Part B).
No. of Feature Columns	15, not including target variable.
No. of Classes	2: returned and not returned.
Target Range	Not applicable because this is a classification problem.
Origin	Publicly available Kaggle e-commerce dataset/competition designed specifically for predicting whether an online purchase will be returned.